

Project Title

RetinaCare

AI-Powered Diabetic Retinopathy Detection

Frontend Implementation

By

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Project Overview

RetinaCare is a web-based interface designed to help users analyze retinal fundus images and identify early signs of diabetic retinopathy.

The frontend connects users to an AI model (via API) that predicts the likelihood of disease progression using uploaded eye images.

The goal is to make early detection accessible, fast, and user-friendly.

Features

- Clean, responsive UI
- Upload retinal images directly from device
- Connects to backend AI model for predictions
- Displays prediction result clearly
- Shows risk level (e.g., mild, moderate, severe)
- Simple navigation and accessible UI
- Mobile-friendly design

Technologies Used

- React / Typescript
- TailwindCSS
- API integration
- GitHub
- Vercel

How It Works (User Flow)

1. User uploads a retinal fundus image
2. The website sends the image to the backend AI model
3. AI analyzes image & predicts DR severity/progression
4. Frontend displays:

Homepage

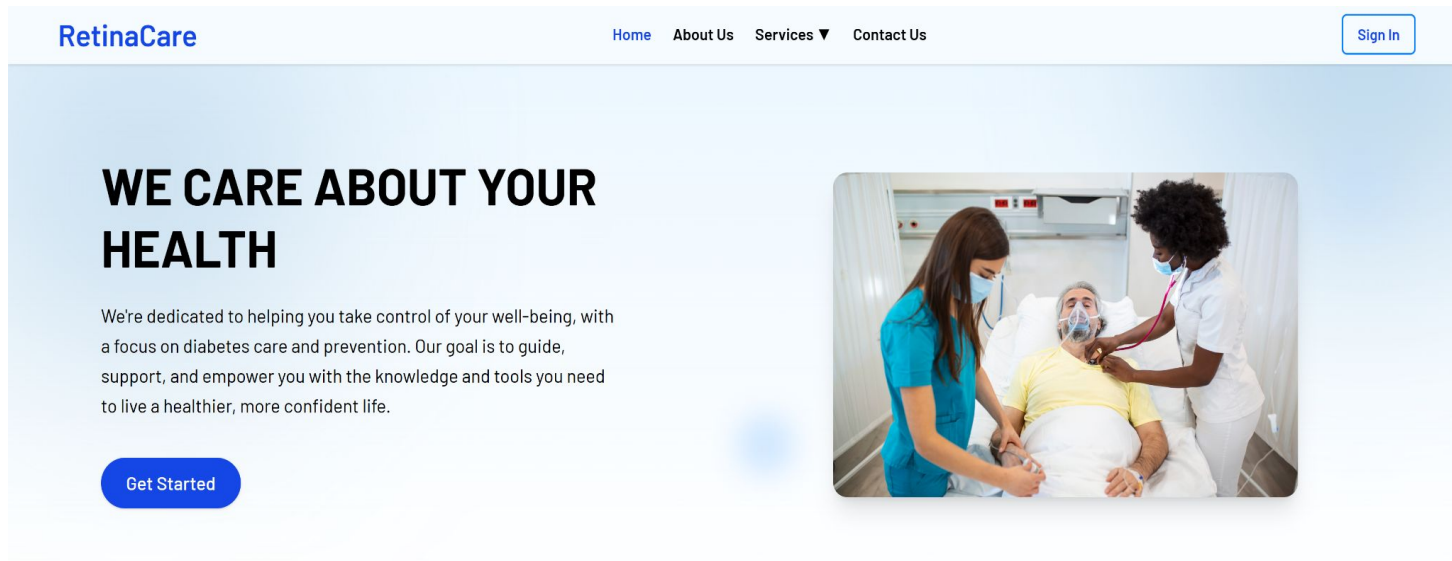
Login/Signup page

Prediction result

Severity level

Confidence score

Desktop HomePage Screenshot



LoginPage Screenshot

RetinaCare
Log into your Account
Let's get started

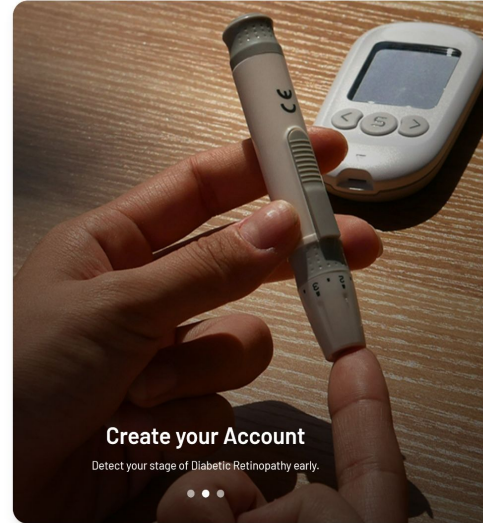
E-mail

Password

[Sign In](#)

 Continue with google

[Don't have an account? Sign Up](#)



SignUpPage Screenshot

RetinaCare
Create your Account
Let's get started

Full name

E-mail

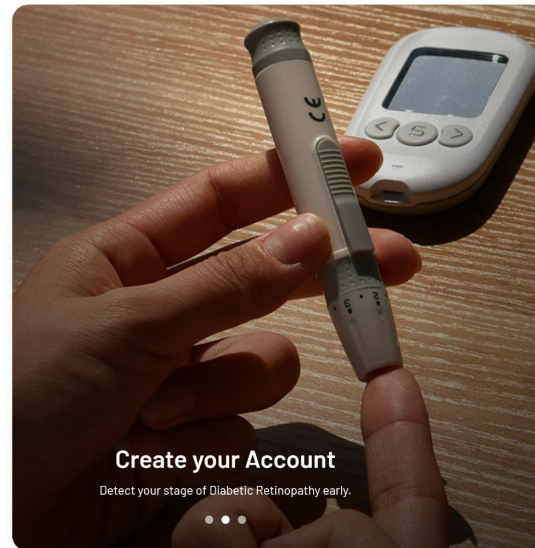
Password

Confirm Password

[Sign Up](#)

[Continue with google](#)

Already have an account? [Sign In](#)



Create your Account

Detect your stage of Diabetic Retinopathy early.

PredictionPage Screenshot

RetinaCare

Sign Out

Patient Data

Fundus Retinal Image

Click or drag to upload

JPG or PNG (Max 10MB)

HbA1c Level (%)

Duration of Diabetes (years)

Systolic BP (mmHg)

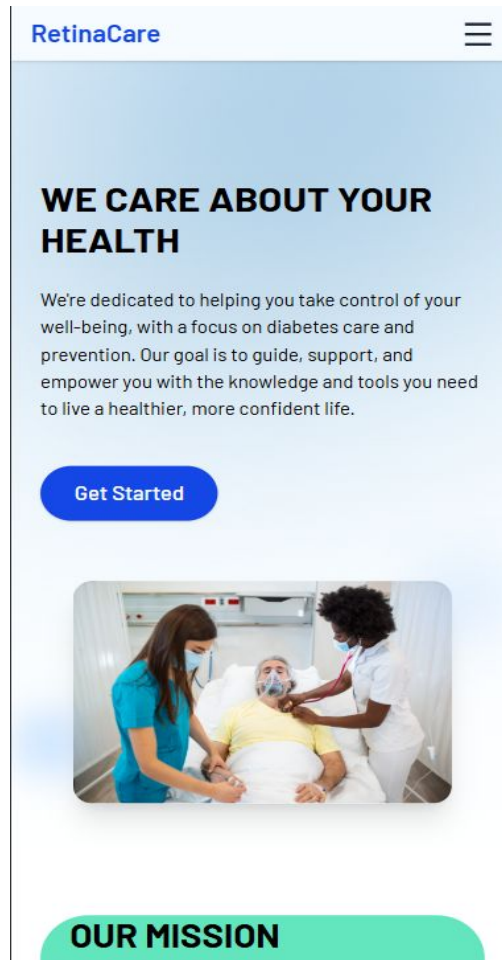
Analyze & Predict Risk

Analysis Report

No prediction generated yet

Complete the patient form and upload an image to receive AI-assisted analysis.

Mobile Screen Screenshot



Challenges Faced

- Connecting frontend to a machine learning API
- Handling image uploads and preview
- Designing a clean medical-style UI
- Ensuring mobile responsiveness
- Understanding medical terminology for proper UI labeling

Lessons Learned

- Improved UI/UX design skills
- Experience with API integration
- Understanding how to visualize AI outputs
- Better knowledge of frontend structuring and file management
- Building responsive and accessible web pages

Conclusion

RetinaCare helps support early detection of diabetic retinopathy using AI.
This frontend ensures the platform is accessible, fast, and easy to use for all users.

Thank you.